Smart Adaptive Load-Aware Solar Battery Charging System

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***Abstract*—The Smart Adaptive Load-Aware Solar Battery Charging System presented in this study is intended for universal EV charging applications. This suggested system combines Maximum Power Point Tracking (MPPT) with a buck-boost converter to get maximum solar energy in a range of several irradiance conditions. In contrast to traditional solar chargers, the system considers multiple factors such as adaptive current control depending on battery health and other operating conditions, state-of-charge (SoC) estimation using Coulomb counting method, and battery identification through capacity. The simulation and hardware results try to bring in a system that is efficient in power conversion and identifies multiple configurations of battery loads, then charges according to it.**

***Index Terms*—Solar charger, MPPT, buck-boost converter, Irradiance, adaptive charging, SoC estimation, EV charging, Coulomb Counting**

I. Introduction

Renewable energy is very much in demand for its abundant availability and sustainability. In recent years, With the emerging EV sector, it is very much required to be integrated with renewable resources. Although they offer sustainable alternatives, Solar-powered charging stations are still not widely considered for charging purposes due to their unpredictable irradiance. So, MPPT is brought in to ensure that, although with the varying range of power, the system gets the maximum amount of power from the source. But even with the consideration of MPPT, there are still several challenges in the system as system parameters such as battery charging stress and ripple effects ocuur in the power stage. So, this system tries to be adaptive control approach that modifies MPPT by bringing in converter working and its various system constraints. The Buck-boost converter is considered because different battery configurations require different amounts of voltage, due to which the converter will have to either step-up or step-down the voltage for the system. Normally, in all EV charging stations, there are separate charging ports for different vehicles as they have varying capacities, This takes a lot of space and infrastructure. Therefore, there is a need of a system that is capable of charging safely, is adaptive to multiple loads and able to scale up to different battery configurations.

II. Literature Review

Early research on photovoltaic systems mainly focused on comparison of different Maximum Power Point Tracking (MPPT) techniques to extract maximum output power from solar arrays [1]. Power electronics is identified as a key contributor in improving energy conversion efficiency and addressing global energy challenges [2].

Residential photovoltaic energy storage systems were introduced to integrate solar generation with battery storage for improved utilization [3]. Later developments emphasized grid-connected photovoltaic generation systems with high step-up voltage gain converters and MPPT integration [4]. Optimization of photovoltaic series-parallel arrays under mismatched conditions was also investigated to improve reliability and power extraction [5].

Real-time MPPT implementation techniques were further developed for practical photovoltaic systems [6]. Theoretical assessments were carried out to evaluate MPPT efficiency with different converter topologies including buck–boost structures [7]. Single-stage grid-connected inverter topologies integrating MPPT were also explored [8].

To improve performance under fast-changing irradiance conditions, modified incremental conductance MPPT algorithms were proposed [9]. Real-time MPPT techniques for mismatched photovoltaic systems were also studied to enhance tracking accuracy and system stability [10].

Although significant advancements have been made in MPPT algorithms and converter topologies, very few works focus on integrating adaptive load-aware charging mechanisms, battery parameter estimation such as State of Charge (SoC), State of Health (SoH), temperature monitoring, and universal multi-configuration battery charging within a single unified system.

A. Literature Summary

Modern research focuses on improved buck–boost converter designs and intelligent MPPT algorithms (sliding-mode, adaptive P&O, fuzzy, ANN). Integration of battery parameters is also not considered much into account which is important for safe and secure charging. Very few focus on multi-load adaptive charger.

B. Research Gap

Lack of a unified charging system, there is no focus on adapting to different battery level configurations. Not considering the battery parameters such as SoC, SoH and temperature.

Very few systems think about preventing the faults that arise in the system. The system should be able to satisfy the power quality, by lowering the ripple and harmonics could the system be more efficient.

C. Motivation

The need for a universal solar charging interface that is capable of safely charging multiple battery types through a single port no matter of any load capacities.

D. Objective

- Develop a universal solar-powered EV charging interface
- Implementing intelligent MPPT tracking that combines system parameters in accounts battery charging current, ripple components, and load mismatch error, and then produces an optimized MPPT value.
- Integrate MPPT with adaptive load control that is basically able to detect any load capacity and arrange the respective battery configuration to it.
- Implement safe charging logic that accounts for the State of Charge(SoC) to prevent overcharging. When charging, if it reaches the threshold limit, the converter should stop producing duty ratio, hence the system will stop charging.

III. Methodology

A. Proposed System-Aware Adaptive Control

To address the limitations of conventional MPPT, such as converter working and battery stress, a system-aware adaptive control approach is proposed. This method tries to bring into converter stability constraints and optimization techniques into the tracking process[9].

1) Multi-Objective MPPT Formulation: The traditional power maximization MPPT objective will be mostly about finding the best point of curve where voltage and current both are balanced to give the best power value. But here it is extended into a multi-objective cost function J to balance power extraction with system safety:

$$J = P - \lambda_1 I_{out}^2 - \lambda_2 \rho^2 - \lambda_3 e_L^2 \tag{1}$$

Where:

- $P = V_{pv} I_{pv}$ represents the ordinary mppt value.
- I_{out} represents the battery charging current, ought to reduce battery stress.
- ρ is the ripple ratio of the inductor current, it is accounted as when after a long duration of time, thermal stress and switching losses will increase so it is used to limit that.
- e_L is the load impedance mismatch error to improve transfer efficiency. Both source and output from converter should have little to no difference in impedence as then it will affect the maximum power generated.
- $\lambda_{1,2,3}$ are weighting coefficients used to prioritize different system constraints.

2) RHP-Zero-Aware Adaptive Step Size: Buck–boost converters show a right half plane (RHP) zero in their transfer function:

$$\omega_z = \frac{R(1-D)^2}{L} \tag{2}$$

When the duty cycle D increases, the RHP zero changes to lower frequencies as they are inversely proportional, limiting the control bandwidth and hence causing instability. To maintain stability, the controller must continuously be able to modify the MPPT step size μ_{eff}:

$$\mu_{eff}(D) = \mu_0 \frac{(1-D)^2}{(1-D_{nom})^2} \tag{3}$$

This mechanism helps in not limiting the controller operating range. It hence helps in using controller for wide operating ranges.

B. State-of-Charge Estimation

State-of-charge is calculated using the Coulomb counting method:

$$SoC = SoC_0 + \frac{1}{C_{rated}} \int I_{bat}(t) dt \tag{4}$$

It combines the battery rated capacity and charge added to calculate SoC. The result represents the percentage of total battery capacity charged, allowing the system to stop charging once the threshold is reached.

IV. Simulation Results and Discussion

The above two figures fig 1 and fig2 show the simulation and the Soc waveform in the MATLAB simulation respectively. The simulation has the components such as; i) Solar panel (with constant irradiance and temperature) ii) An inverting single switch buck-boost converter with a semiconductor device, MOSFET switching. iii) The charge controller, here it's the central processing unit of the system, where the computation of the duty ratio happens for the PWM. With a sawtooth waveform to generate the PWM, which is applied to the MOSFET, and with that switching control in the buck-boost converter, the output voltage. From the above configuration of the system, it is observed that the above system is a closed loop with the output voltage of buck boost converter (or) the battery pack voltage as feedback into the charger controller, with the help of the voltage the PWM duty ratio is updated for every change in output voltage. According to the objective of the system to be achieved now the constant irradiance and temperature in the solar PV array side is replaced with signal generator which replicates the real time solar panel system available with that arrangements the simulation rerunned for a duration of 10 seconds and it is observed that the system gets adapted to the available power to be maintained for the battery pack, the power available divided by the 1C- rate current of the battery pack gives the maximum voltage possible with that the controller again recalculates the PWM duty ratio and PWM is generated from the controller and fed to the buck boost converter and with that requires output voltage is given to the battery pack increases upto its threshold (90%soc) without any input which can be seen in the soc VS time waveform above.

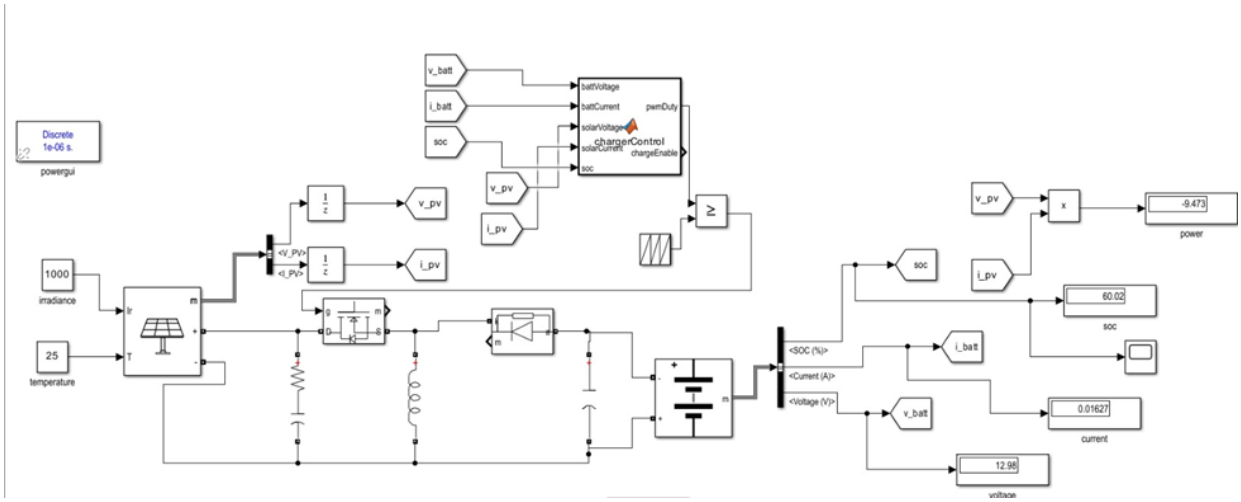


Fig. 1. Simulation of system.

TABLE I
PWM DUTY RATIO COMPARISON FOR DIFFERENT INPUT VOLTAGES AND BATTERY CONFIGURATIONS

Vin (V)	Battery Pack	Target Voltage (V)	PWM Duty Calculated	PWM Duty Observed
12	3S	12.6	0.512	0.519
12	4S	16.8	0.583	0.587
10	3S	12.6	0.557	0.558
10	4S	16.8	0.627	0.631
15	3S	12.6	0.456	0.460
15	4S	16.8	0.528	0.530

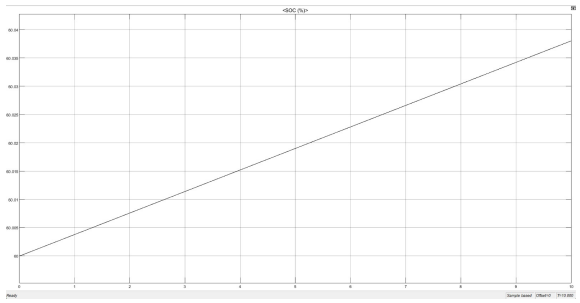


Fig. 2. SoC of battery pack of nominal voltage 12V vs. time.

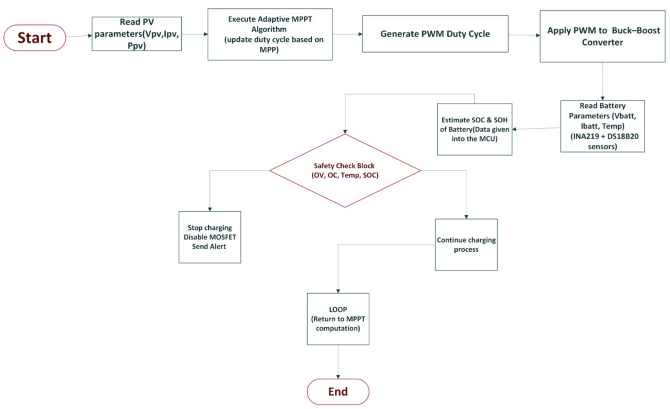


Fig. 3. System architecture and control flow block diagram.

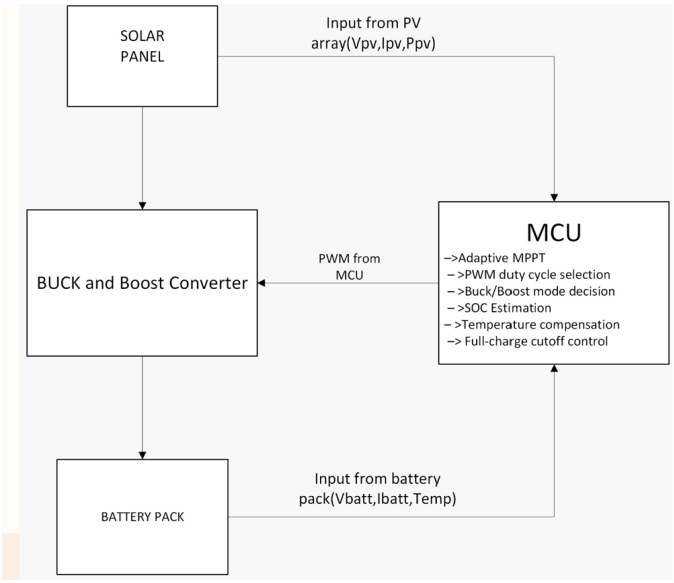


Fig. 4. Block diagram.

V. HARDWARE SETUP

A. Parameter and Rating / Specification

B. Hardware System

The system consists of a buck-boost converter and different battery pack configurations such as 3S and 4S.

TABLE II
HARDWARE SPECIFICATIONS

Component	Specification
Converter Type	Buck–Boost
Controller	Arduino
MOSFET	60V, 30A
Inductor	0.5 mH
Output Capacitor	330µF, 470µF

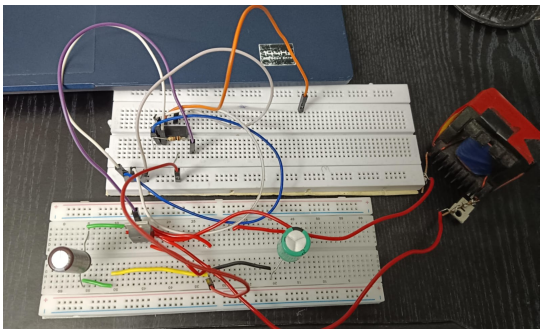


Fig. 5. Hardware system.

VI. HARDWARE RESULTS AND DISCUSSION

Experimental results show the varying pwm given to the battery as it changes multiple times due to the step size ensuring all the system paramters are accounted while charging and ensuring efficient MPPT operation.

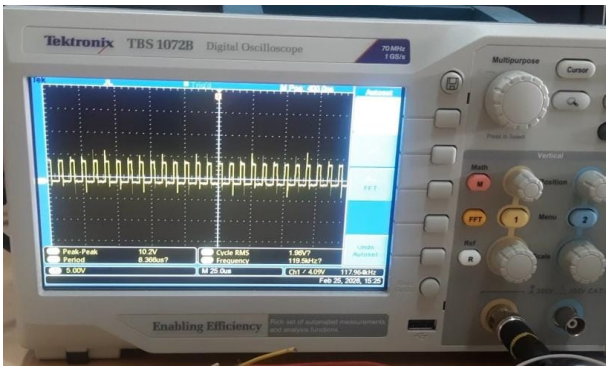


Fig. 6. Observed hardware output waveforms.

VII. CONCLUSION AND FUTURE SCOPE

A Smart Adaptive Load-Aware Solar Battery Charging System has been successfully designed and validated. The system integrates MPPT with adaptive load-aware charging logic. During each control cycle, the controller evaluates the cost function and determines the gradient direction that increases the objective function. The duty cycle of the converter is then updated according to the adaptive step size rule. This process allows the converter to gradually converge toward an operating point that satisfies both power extraction and system constraints. Here, the system is designed of two battery configurations that is 3S and 4S, the capacities are detected and then with respect to that the converter decides to buck or boost with differing duty ratios. The system calculates SoC, also accounts battery parameters for safe charging. The future scope includes the estimation of battery health based on AI by integrating machine learning techniques and bi-directional topology in converters. More and more IoT frameworks can be used for fault detection, monitoring and analysis.

VIII. REFERENCES

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